SEP 2025 **DATA STURCTURES IN C** **80MARKS**

SECTION -A

ANSWER ANY TEN QUESTIONS. **10*2=20**

1. List the advatages of Pointer.
2. Mention the types of memory allocation techniques.
3. Define Datastructure and mention ite types.
4. List the different modes to open the file.
5. Define Recursion. Give an Example.
6. What is Stack? List its applications.
7. Mention the types of Queue.
8. Define height and level of Queue.
9. What is sorting? List the sorting techniques.
10. Defferentiate between linear search and binary search.
11. What is complete binary tree.
12. Mention the types of Traversal.

SECTION-B

Answer anty six questions? **4*5=20**

13. State the difference b/w static and dynamic memory allocation.
14. Explain the file error handling functions.
15. Write a Porgram in "C" to sort elements in ascending order using bubble sort.
16. Explain double ended queue.
17. Explian different typesof linked list with examples.

SECTION-C

Answer any three Questions 4*10=40

18. (a) Differentiate between malloc() and calloc().
    (b) What is Pointer? Write the advantages and disadvantages of Pointer.
19. (a) Write a "c" program to generate "n" Fibonacci numbers using recursive function.
    (b) What is file? Explain any five file I/O functions.
20. (a) Sort the following elements using selection sort method.
    15, 8, 7, 19, 1, 5, 7
    (b) Write "C" Program to search on element using Binary search technique.
21. (a) Convert the following infix expression to post fix and prefix form.
    $a \quad b \quad c - d + e + f/g.$
    (b)Write a "C" Program to demonstrate the working of simple queue.
22. (a) Define the following terms:
    i) Root ii) Node iii) Subtree iv) Sibling v) Path
    (b) Define Linked List. Explain its Components with neat diagram.